WNBA Win Probability Model

Model Validation & Methodology Report

Generated July 23, 2026 — Luke York

Overview

This model predicts WNBA game win probabilities using a logistic regression trained on team form, scoring margin, an Elo rating system, and rest days. It is validated using walk-forward cross-validation across multiple seasons, benchmarked against naive baselines, and evaluated for both accuracy and probability calibration, since a model intended to inform real decisions needs to be right about its confidence level, not just its picks.

Methodology

Features: rolling win percentage (10-game window), rolling scoring margin, Elo rating (margin-of-victory weighted, with season-to-season regression toward the mean), and rest days between games.

Validation: walk-forward cross-validation — for each test season, the model is trained only on seasons that came strictly before it, so no future information leaks into training. This mirrors how the model would actually be used: always predicting forward in time.

Baselines: compared against (1) always picking the home team, and (2) Elo rating alone, to isolate how much the additional features actually contribute beyond a single strong predictor.

Walk-Forward Validation Results

Test Season	Home-Always-Wins	Elo-Only	Full Model
2023.0000	0.5299	0.6604	0.6679
2024.0000	0.5402	0.6667	0.6782
2025.0000	0.5683	0.6698	0.6667

The full model beats the home-team baseline by 10-13 percentage points in every tested season, and modestly outperforms Elo alone in two of three seasons — indicating Elo is the dominant predictive signal, with the additional features (form, margin, rest) providing a smaller, not fully consistent, incremental improvement.

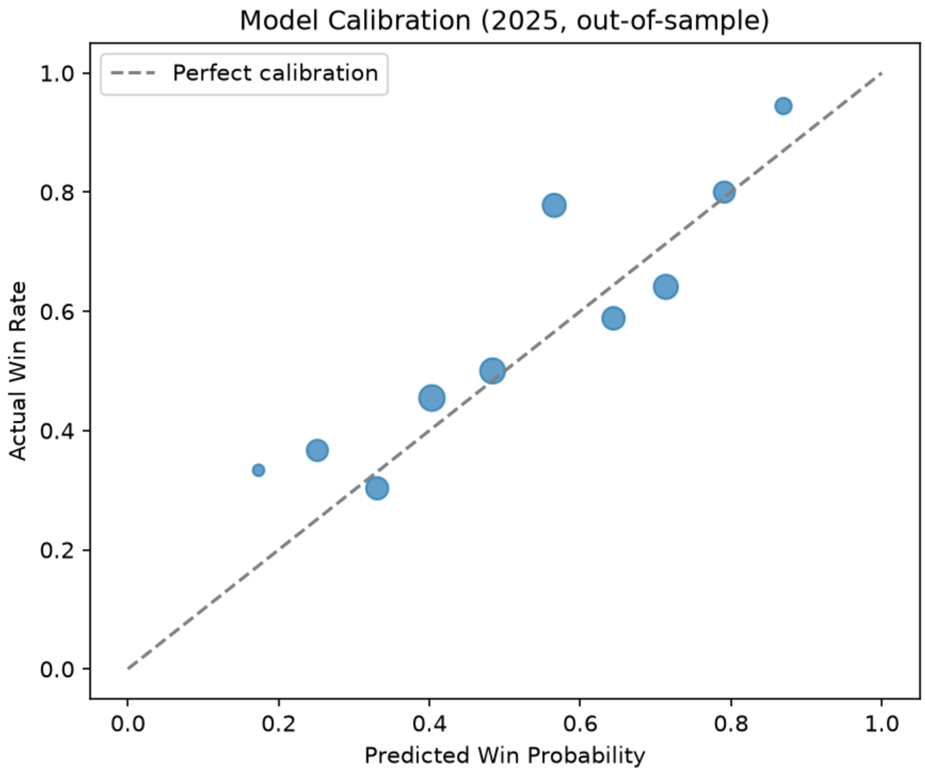
Out-of-Sample Metrics (2025)

Accuracy	Log Loss	Brier Score

0.6667	0.6241	0.2179

Log loss and Brier score are both meaningfully better than a pure-guessing baseline (0.693 log loss, 0.25 Brier score for always guessing 50%), indicating the model's confidence levels carry real information, not just its win/loss picks.

Calibration



Points near the dashed diagonal indicate well-calibrated predictions (e.g. games predicted at 80% confidence actually won about 80% of the time). Most bins track closely; a few bins with small sample sizes (under 20 games) show larger deviations, which reflects limited data in those confidence ranges rather than a systematic flaw in the model.

Limitations & Honest Caveats

• Calibration in several confidence bins is based on fewer than 30 games; more historical seasons would improve confidence in those estimates.

• The model does not account for player-level information (injuries, star player rest/availability), which materially affects real WNBA outcomes.

• Elo ratings regress partially toward the mean between seasons, which is a simplification of true roster turnover.

• This model reflects historical patterns and is not a guarantee of future performance. Any use for betting purposes carries real financial risk.